Ss 1A

```
vasisht-k-gaddale@vasisht-k-gaddale-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/PES1UG24CS521-pes-vcs$ make clean
make test_objects
./test_objects
rm -f pes test_objects test_tree object.o tree.o index.o commit.o pes.o test_objects.o test_tree.o
rm -rf .pes
gcc -Wall -Wextra -O2 -c test_objects.c -o test_objects.o
gcc -Wall -Wextra -O2 -c object.c -o object.o
object.c: In function ‘object_write’:
object.c:99:47: warning: ‘.tmp’ directive output may be truncated writing 4 bytes into a region of size between 1 and 512 [-Wformat-truncation=]
   99 |     snprintf(temp_path, sizeof(temp_path), "%s.tmp", path);
      |                                               ^~~~
In file included from /usr/include/stdio.h:970,
                 from pes.h:8,
                 from object.c:3:
In function ‘snprintf’,
    inlined from ‘object_write’ at object.c:99:5:
/usr/include/x86_64-linux-gnu/bits/stdio2.h:68:10: note: ‘__builtin___snprintf_chk’ output between 5 and 516 bytes into a destination of size 512
   68 |   return __builtin___snprintf_chk (__s, __n, __USE_FORTIFY_LEVEL - 1,
      |          ^~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
   69 |                                    __glibc_objsize (__s), __fmt,
      |                                    ~~~~~~~~~~~~~~~~~~~~~~~~~~~~
   70 |                                    __va_arg_pack ());
      |                                    ~~~~~~~~~~~~~~~~~
gcc -o test_objects test_objects.o object.o -lcrypto
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

vasisht-k-gaddale@vasisht-k-gaddale-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/PES1UG24CS521-pes-vcs$
```

SS 1B

```
vasisht-k-gaddale@vasisht-k-gaddale-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/PES1UG24CS521-pes-vcs$ find .pes/objects -type f
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
vasisht-k-gaddale@vasisht-k-gaddale-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/PES1UG24CS521-pes-vcs$
```

SS 2A

```
find .pes/o
```

Then:

```
vasisht-k-gaddale@vasisht-k-gaddale-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/PES1UG24CS521-pes-vcs$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization
```

SS 2B

```
vasisht-k-gaddale@vasisht-k-gaddale-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/PES1UG24CS521-pes-vcs$ find .pes/objects -type f
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
vasisht-k-gaddale@vasisht-k-gaddale-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/PES1UG24CS521-pes-vcs$ xxd $(find .pes/objects -type f | head -1) | head -20
00000000: 626c 6f62 2031 3600 4865 6c6c 6f2c 2050  blob 16.Hello, P
00000010: 4553 2d56 4353 210a                      ES-VCS!.
vasisht-k-gaddale@vasisht-k-gaddale-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/PES1UG24CS521-pes-vcs$
```

SS 3A

```
vasisht-k-gaddale@vasisht-k-gaddale-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/PES1UG24CS521-pes-vcs$ ./pes init
echo "hello" > a.txt
./pes add a.txt
./pes status
Initialized empty PES repository in .pes/
Staged changes:
  staged:     a.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  tree.c
  untracked:  test_sequence.sh
  untracked:  index.h
  untracked:  commit.h
  untracked:  file.txt
  untracked:  tree.h
  untracked:  index.c
  untracked:  test_tree.c
  untracked:  test_objects.c
  untracked:  pes.c
  untracked:  commit.c
  untracked:  Makefile
  untracked:  object.c
  untracked:  .gitignore
  untracked:  pes.h
  untracked:  README.md

vasisht-k-gaddale@vasisht-k-gaddale-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/PES1UG24CS521-pes-vcs$
```

SS 3B

```
./pes add a.txt
./pes status
Initialized empty PES repository in .pes/
Staged changes:
  staged:     a.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  tree.c
  untracked:  test_sequence.sh
  untracked:  index.h
  untracked:  commit.h
  untracked:  file.txt
  untracked:  tree.h
  untracked:  index.c
  untracked:  test_tree.c
  untracked:  test_objects.c
  untracked:  pes.c
  untracked:  commit.c
  untracked:  Makefile
  untracked:  object.c
  untracked:  .gitignore
  untracked:  pes.h
  untracked:  README.md

vasisht-k-gaddale@vasisht-k-gaddale-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/PES1UG24CS521-pes-vcs$ cat .pes/index
100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776681777 6 a.txt
vasisht-k-gaddale@vasisht-k-gaddale-Victus-by-HP-Gaming-Laptop-15-fa0xxx:~/PES1UG24CS521-pes-vcs$
```

SS 4A

```
commit 5dfb36ccda68e3cc58a8660d3b763ae1a9a6234b33ad44c70295ce93c46f7589
Author: PES User <pes@localhost>
Date:   1776426791

    Add bye file

commit a522ca894927e0196fb042291093a880e2d8f47708fd156a67ae240698157ffe
Author: PES User <pes@localhost>
Date:   1776426763

    Add worldile

commit e9b16c9c971f1492ebf4f0e94e6987e85d2d6dfc3f744312e55a72fbf3cb196d
Author: PES User <pes@localhost>
Date:   1776426686

    Initial committial commiQ
```

SS 4B

```
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/2c/f8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4
.pes/objects/48/1407f7237f7bd749dca31e2d8422bafe211741d5556f007e0faf821ffc7179
.pes/objects/5d/fb36ccda68e3cc58a8660d3b763ae1a9a6234b33ad44c70295ce93c46f7589
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/7d/c3b9d4cee22f645e501f8aaa822f6777b3198434adc74491632478799e9e8b
.pes/objects/92/19086282c09f0db9088484e1ef89f1b05ecfb420a8784a6de327022771fbec
.pes/objects/a5/22ca894927e0196fb042291093a880e2d8f47708fd156a67ae240698157ffe
.pes/objects/e0/0c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f
.pes/objects/e9/b16c9c971f1492ebf4f0e94e6987e85d2d6dfc3f744312e55a72fbf3cb196d
.pes/objects/fe/bed41618fe89b49ea9b82684254a1e74aa7f80fdba38fc84b592ccca955267
.pes/refs/heads/main
```

SS 4C

```
5dfb36ccda68e3cc58a8660d3b763ae1a9a6234b33ad44c70295ce93c46f7589
```

```
ref: refs/heads/main
```

Analysis

Q5.1: Implementing pes checkout <branch>

A branch in PES is a file located in .pes/refs/heads/ that stores a commit hash. Creating a branch is equivalent to creating this file.

Steps to implement pes checkout <branch>:
1. Check if .pes/refs/heads/<branch> exists.
2. Update .pes/HEAD to:
   ref: refs/heads/<branch>
3. Read the commit hash from the branch file.
4. Load the corresponding commit and retrieve its tree.

Update the Working Directory:
- Add or remove files to match the commit
- Restore file contents from blob objects
- Recreate directory structure if necessary

Update Index:
- Update .pes/index to reflect the checked-out state

Complexity:
- Handle uncommitted changes safely
- Prevent accidental overwrites
- Manage file deletions correctly
- Maintain consistency between HEAD, index, and working directory

Q5.2: Detecting a Dirty Working Directory Conflict

Step 1: Check for Modified Files
- Use stat() to check file size and modification time
- If different from index → file is modified

Step 2: Compare with Target Branch
- Compare index hash vs target tree hash

Conflict occurs if:
- File modified in working directory AND
- File differs in target branch

Result:
Checkout must refuse to prevent data loss.

Q5.3: Detached HEAD State

- HEAD contains commit hash instead of branch reference
- Commits are not attached to any branch

Recovery:
- pes branch <new-branch>
- Or manually assign branch to commit

Q6.1: Deleting Unreachable Objects

Algorithm: Mark-and-Sweep

Step 1: Mark
- Start from branch heads
- Traverse commits, trees, blobs

Step 2: Sweep
- Delete objects not reachable

Use Hash Set for efficiency

Q6.2: GC and Race Conditions

Problem:
GC during commit may delete new objects

Solution:
- Atomic writes (temp + rename)
- Grace period
- Locks and reflogs